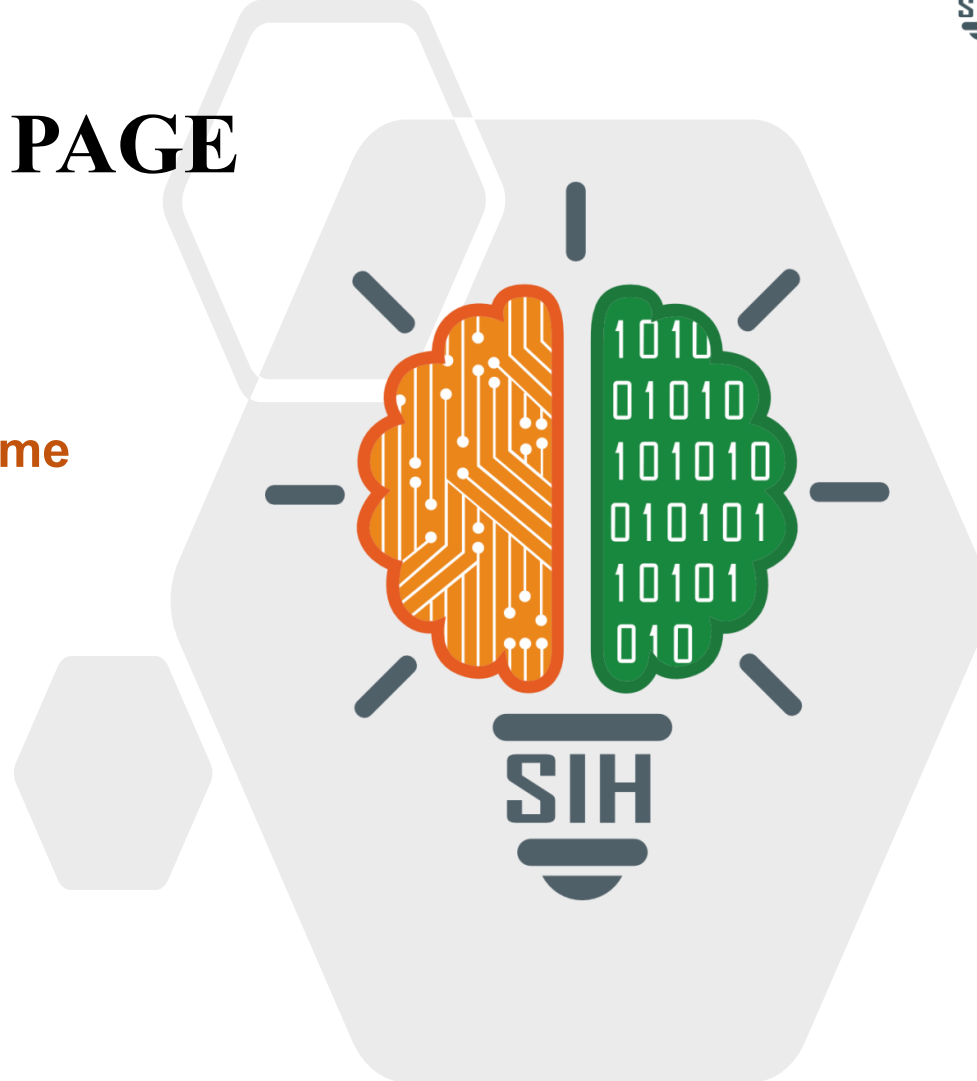


SMART INDIA HACKATHON 2026



TITLE PAGE

- Problem Statement ID – **26092**
- Problem Statement Title – **AI-Driven Scheme Matching for Marginalized Entrepreneurs**
- Theme – **Miscellaneous**
- PS Category – **Software**
- Team ID – **PPSU/SIH2026/08**
- Team Name – **Error 429**



Proposed Solution (Describe your Idea/Solution/Prototype)



One profile, and a citizen instantly sees every central and state scheme they qualify for — then applies with auto-filled forms and a ready-to-submit document kit.



Detailed explanation of the proposed solution

- **One profile** — every scheme scored, with the reason shown
- **Digital vault** — upload a document once, reuse it forever
- **Browser OCR** — the phone reads it, AI then verifies it
- **Auto-fill + Smart Kit** — one indexed, submission-ready PDF
- **Live tracking** — status for the citizen, portal for the admin



How it addresses the problem

- **Discovery gap** — matching is proactive, no searching needed
- **“Am I eligible?”** — criteria met and missing, not a yes/no
- **Rejections** — readiness checked before anything is filed
- **Life events** — marriage, a child or job loss unlock schemes
- **Middlemen** — the core platform costs the citizen nothing



Innovation and uniqueness of the solution

- **On-device OCR** — the document never has to leave the phone
- **AI gatekeeping** — confirms it is the right document first
- **Smart Document Kit** — cover page, index and ordered PDF
- **Explainable score** — criteria shown, never a black box
- **Readiness ranking** — exactly what to fix first, in order

TECHNICAL APPROACH

Technologies to be used (e.g. programming languages, frameworks, hardware)



Frontend

Next.js 14 App Router, React 18, TypeScript, Tailwind CSS, Radix UI



Backend & Data

Supabase — PostgreSQL, Auth, Storage, Realtime; Row-Level Security



AI Layer

Groq LLaMA-3.3-70B — document classification, extraction, reasoning



OCR & Documents

Tesseract.js WASM OCR in Hindi and English; pdf-lib kit builder



Platform

Vercel, Turborepo, Razorpay, SMS and e-mail, AES-256-GCM encryption

Methodology and process for implementation (Flow Charts/Images/ working prototype)



Profile

One-time profile, consent captured



Match

Every scheme scored, with reasons



Upload

Browser OCR reads the document



Verify

AI confirms type, extracts fields



Apply

Auto-filled form + Smart Kit PDF



Track

Status updates and admin review



Working prototype already runs end-to-end — 36 API routes, 60 React components, ~21,000 lines of TypeScript; browser OCR, AI verification and PDF kit generation are functional.

FEASIBILITY AND VIABILITY



Analysis of the feasibility of the idea

- **Built** — the full journey already works end to end today
- **Modular** — scheme rules, documents and workflows can be updated independently
- **Explainable** — every recommendation can show what matched and what is missing
- **Unblocked** — the MVP runs on published data, no API gate
- **Maintainable** — multi languages, one framework, managed infra



Potential challenges and risks

Scheme rules change constantly across thousands of central and state schemes

OCR struggles with faded, handwritten or regional-language documents

Aadhaar-class documents carry a real privacy and legal duty

Low digital literacy in exactly the households that need this most

Citizens hesitate to upload identity papers to an unfamiliar platform



Strategies for overcoming these challenges

A curated index with versioning and citizen-reported corrections

Two-layer OCR — Tesseract plus AI validation, manual entry as fallback

Encrypted fields, per-user access rules and short-lived document links

Hindi and English interface, assisted mode for CSC and NGO operators

Clear consent, no data sale and delete-on-request to earn that trust

IMPACT AND BENEFITS

 **5,000+**

central and state schemes
brought into one place

 **1 profile**

unlocks every scheme the
citizen qualifies for

 **Minutes**

from sign-up to a
submission-ready kit

 **₹0**

cost to the citizen for
the core service

Potential impact on the target audience



Students



Farmers



Women



Seniors



PwD

- **Discovery** — days of searching scattered portals become minutes
- **Fewer rejections** — papers are checked for fit before anything is filed
- **Second chances** — life-event alerts reach people who stopped looking
- **Reach** — an operator can run the same flow for a rural citizen



Benefits of the solution (social, economic, environmental, etc.)



Social — entitlements reach the people they were written for, with less dependence on paid middlemen



Economic — households recover benefits they already qualify for; cost per extra user stays near zero



Environmental — a paperless vault and digital kit cut printing and repeated trips to government offices



Governance — administrators get live analytics on demand, backlog, approval rates and document gaps



Inclusion — one Hindi and English flow serves the citizen directly or through an assisting operator



Details / Links of the reference and research work



Government sources and scheme data

- myScheme — national scheme discovery platform
myscheme.gov.in
- National Scholarship Portal
scholarships.gov.in
- DBT Bharat — Direct Benefit Transfer Mission
dbtbharat.gov.in
- DigiLocker — issued document framework
digilocker.gov.in
- India.gov.in — schemes and services directory
india.gov.in/schemes
- UIDAI — Aadhaar data protection guidance
uidai.gov.in



Technical research and documentation

- Tesseract.js — in-browser OCR engine
tesseract.projectnaptha.com
- Groq — LLaMA-3.3-70B inference API
console.groq.com/docs
- Supabase — Row-Level Security guide
supabase.com/docs
- pdf-lib — PDF creation in JavaScript
pdf-lib.js.org
- Next.js 14 — App Router documentation
nextjs.org/docs
- Vercel — serverless deployment platform
vercel.com/docs



Video walkthrough
youtu.be/K8FrSU-rOlc



Live demo
sara-sahayta.vercel.app